

Integration — Lesson 23 Test: The Trapezoidal Rule

10 questions · show your work in the box · calculators allowed · include UNITS where they apply

Name: _____ Date: _____ Score: _____ / 10

1. A trapezoid standing on the x-axis has a left side of height 5, a right side of height 9, and width 4. Find its area. 1 pt

2. A function has $f(2) = 6$ and $f(6) = 10$. Use ONE trapezoid to estimate $\int_2^6 f(x) \, dx$. 1 pt

3. Use TWO trapezoids ($\Delta x = 1$) to estimate $\int_0^2 x^2 \, dx$. (Heights: $f(0) = 0$, $f(1) = 1$, $f(2) = 4$.) 1 pt
Then say whether your estimate is bigger or smaller than the true value $8/3 \approx 2.667$.

4. For a function on $[0, 4]$ with equal-width slices, the left Riemann sum is 12 and the right Riemann sum is 20. What is the trapezoid sum? Name the rule you used, and state the condition it requires. 1 pt

5. Use THREE trapezoids ($\Delta x = 1$) to estimate $\int_0^3 x^2 dx$, then compute the true value with the FTC. By how much is the estimate off? 1 pt

6. A runner's velocity (m/s) is recorded at uneven times: 1 pt

t (s)	0	2	6	9
v(t) (m/s)	5	11	15	9

Use a trapezoidal sum with the three subintervals shown to estimate $\int_0^9 v(t) dt$, with units.

7. Water flows into a barrel at rate $R(t)$ gallons per minute:

1 pt

t (min)	0	5	8	12
$R(t)$ (gal/min)	4	10	14	6

Use a trapezoidal sum with the three subintervals shown to estimate the total water added from $t = 0$ to $t = 12$, with units.

8. A function g is concave UP on $[1, 7]$. Will a trapezoidal sum for $\int_1^7 g(x) dx$ overestimate or underestimate the true value? Justify your answer using chords. 1 pt

9. For $\int_0^2 x^3 dx$ with two slices ($\Delta x = 1$): compute the trapezoid sum and the midpoint sum, then compute the TRUE value with the FTC. Which estimate is closer? (Heights: $f(0) = 0$, $f(0.5) = 0.125$, $f(1) = 1$, $f(1.5) = 3.375$, $f(2) = 8$.) 1 pt

10. Cars pass a checkpoint at rate $R(t)$ cars per hour:

1 pt

t (hours)	0	3	7	10
R(t) (cars/hour)	40	60	50	30

(a) Use a trapezoidal sum with the three subintervals shown to estimate $\int_0^{10} R(t) dt$. (b) Using correct units, explain what your answer means.
